



THE OVERDOSE

Harm in a Wired Hospital

How Medical Tech Gave a Patient a Massive Overdose

Pablo Garcia went to the hospital feeling fine.
Then the hospital made him very sick.

The nurses and doctors summoned to the hospital room of 16-year-old Pablo Garcia early on the morning of July 27, 2013, knew something was terribly wrong. Just past midnight, Pablo had complained of numbness and tingling all over his body. Two hours later, the tingling had grown worse.

Although Pablo had a dangerous illness—a rare genetic disease called NEMO syndrome that leads to a lifetime of frequent infections and bowel inflammation—his admission to the University of California, San Francisco Medical Center’s Benioff Children’s Hospital had been for a routine colonoscopy, to evaluate a polyp and an area of intestinal narrowing.

At 9 o’clock that night, Pablo took all his evening medications, including steroids to tamp down his dysfunctional immune system and antibiotics to stave off infections. When he started complaining of the tingling, Brooke Levitt, his nurse for the night, wondered whether his symptoms had something to do with GoLYTELY, the nasty bowel-cleansing solution he had been gulping down all evening to prepare for the procedure. Or perhaps he was reacting to the antinausea pills he had taken to keep the GoLYTELY down.

Levitt’s supervising nurse was stumped, too, so they summoned the chief resident in pediatrics, who was on call that night. When the physician arrived in the room, he spoke to and examined the patient, who was anxious, mildly confused, and still complaining of being “numb all over.”

He opened Pablo’s electronic medical record and searched the medication list for clues that might explain the unusual symptoms.

At first, he was perplexed. But then he noticed something that stopped him cold. Six hours earlier, Levitt had given the patient not one Septra pill—a tried-and-true antibiotic used principally for urinary and skin infections—but 38½ of them.

Levitt recalls that moment as the worst of her life. “Wait, look at this Septra dose,” the resident said to her. “This is a huge dose. Oh my God, did you *give* this dose?”

“Oh my God,” she said. “I did.”

The doctor picked up the phone and called San Francisco’s poison control center. No one at the center had ever heard of an accidental overdose this large—for Septra or any other antibiotic, for that matter—and nothing close had ever been reported in the medical literature. The toxicology expert

there told the panicked clinicians that there wasn't much they could do other than monitor the patient closely.

As a precaution, the hospital's rapid response team was summoned to the room. Pablo's mother, Blanca, who had been with her younger son, hospitalized one floor up at UCSF for a severe skin infection (he, too, suffers from NEMO syndrome), began a vigil by Pablo's bedside. "I phoned my sister, and we prayed together," she later recalled.

At 5:32 a.m., Brooke Levitt heard a scream coming from Pablo's room. It was Blanca Garcia. A few seconds earlier, her son had sat bolt upright in bed, yelled out "Mom!" then flopped backward. Levitt sprinted to the room, and when she got there, Pablo's head was snapping back and forth, teeth clenched, back arched, extremities thrashing. He was having a grand mal seizure. Moments later, just as the Code Blue team arrived, the teenager stopped breathing.

"I thought, what if I killed him?" Levitt told me months later, wiping away tears. "If he had a seizure, I'm wondering if that's going to be the end of it. . . I'm trying to hold it together, but I'm in shock the whole time. I just felt immensely guilty."



The night that Pablo Garcia was given a 39-fold overdose of a routine antibiotic offers a cautionary tale that cannot be ignored.

To appreciate how one of the nation's best hospitals—*US News & World Report* regularly ranks UCSF among the top 10—could give a patient a 39-fold overdose of a common antibiotic, one first needs to understand how medicines were ordered and administered in hospitals as recently as a few years ago, before the system went digital.

Pablo Garcia had been taking one double-strength Septra tablet twice a day at home to prevent his frequent skin infections. In the old paper-based medication ordering system, the admitting doctor would have written "Septra 1 ds bid" (using the Latin abbreviation for "twice a day") in the "Doctors' Orders" section of a paper chart, a stack of sheets contained within a plastic three-ring binder.

The physician would have turned a colored wheel on the side of the binder to green, signaling to the ward clerk that there was an order to be "taken

off.” The clerk would then have faxed the order sheet to the pharmacy, where a pharmacist would have read it, signified his approval by initialing the page, and handed the copy to a technician, who would have grabbed a big bottle of Septra pills from a shelf. The tech would then have poured out the pill, or perhaps a few days’ worth of pills, and put them in a bag or cup, which would have then been delivered to the patient’s floor by a runner or a pneumatic tube system.

Once the pills arrived on the floor, at the appropriate time the patient’s nurse would have read the order (manually transcribed from the doctor’s order sheet to the nurse’s Medication Administration Record) and entered the teenager’s room pushing a wheeled cart similar to the ones used by airline flight attendants. After opening the patient’s drawer in the cart, the nurse would have removed the medication, and others due to be given at the same time, watched the patient take the pill, and placed her signature next to the time and dose on her record.



Believe it or not, I’ve shortened this description—time and motion studies have identified as many as 50 steps between the moment the doctor wrote the order and the moment the nurse finally administered the medication. But even in simplified form, you can see why the old system was hugely error-prone. A study from the pen-and-paper era showed that 1 in 15 hospitalized patients suffered from an adverse drug event, often due to medication errors. A 2010 study (using data collected during the pre-digital era) estimated the yearly cost of medication errors in U.S. hospitals at \$21 billion.

Those of us who worked in this Rube Goldberg system—and witnessed the harms it caused—anxiously awaited the arrival of computers to plug its leaks. Computerized ordering would make a doctor's handwriting as irrelevant as scratches on a record album. Computerized decision support would alert the doctor or pharmacist that the patient was allergic to the medication being ordered, or that two medications might interact dangerously. A pharmacy robot could ensure that the right medication was pulled off the shelf, and that the dose was measured with a jeweler's precision. And a bar-coding system would render the final leg in this relay race flawless, since it would signal the nurse if she had grabbed the wrong medication or was in the wrong patient's room.

Of course, it was natural for doctors, nurses and pharmacists to expect that, once computers entered our complex, chaotic and often dangerous world, they would make things better. After all, in our off-duty lives we are so thoroughly used to taking out our iPhones, downloading an app, and off we go.

But we're learning that the magic of information technology, so familiar to us in the consumer world that it nearly seems "normal," is far more elusive in the world of medicine.

Though computers can and do improve patient safety in many ways, the case of Pablo Garcia vividly illustrates that, even in one of the world's best hospitals, filled with well-trained, careful and caring doctors, nurses and pharmacists, technology can cause breathtaking errors.

This one began when a young physician went to an electronic health record and set a process in motion that never could have happened in the age of paper.



At around noon on a cool July day in San Francisco, Jenny Lucca, a pediatrics resident at UCSF, began the process of admitting Pablo Garcia, whose rare genetic disease had led to bouts of gastrointestinal bleeding and abdominal pain. He needed further evaluation with an elective colonoscopy, and this was a scheduled admission to perform the test and act on its results.

After speaking to Pablo and his mother and examining the young patient, Lucca clicked into the physicians' orders section in the electronic health record. Pablo was on about 15 different medications. Lucca ordered his usual immunosuppressive pills, the liquid bowel-cleansing prep for the colonoscopy (the famously vile GoLYTELY) and his monthly infusion of immunoglobulins.

Then she came to the Septra, an antibiotic that the teenager had been taking for years to prevent recurrent skin and lung infections. The usual dose of Septra for all but the smallest children is one double-strength pill twice daily, and that is what Pablo was taking at home.

In the precomputer days, of course, Lucca would have written simply to continue the Septra, twice daily, on the physician's order sheet.

But UCSF Medical Center, where I work as a physician, had not relied on paper for years. It had been over a decade since doctors and nurses wrote their daily notes on paper, and all of the orders had been electronic for nearly two years. Lucca, as a young physician, had never experienced a medical profession built on a backbone of paper documentation. She was of a generation of digital natives, for whom the use of the computer was natural and expected. After arriving in San Francisco, Lucca took the required 10 hours of computer training, and UCSF's system, built by Epic of Verona, Wisconsin—the same one she had used in medical school—made the learning curve far less steep than it might have been.

The medical center installed its first hospital-wide computer system in 2000. We switched to Epic, the market leader, in 2012 after an unhappy decade with General Electric's problem-ridden EHR system. Our implementation of Epic, like all such implementations, had its share of hiccups. Some departments didn't send out bills for weeks, some medications and lab tests were overlooked, and a few patients fell off the hospital's radar screen for brief periods. Like new homeowners, the IT department had a "punch list" of hundreds of items to be fixed or modified, and they spent much of the first year after implementation methodically going through it, checking off items.

But now, on the date of Pablo Garcia's admission, 13 months after UCSF's Epic installation, the system was running smoothly. And there was good evidence that it was meeting its goals: doctors' and nurses' notes were now legible; thousands of medication errors had been intercepted by the bar-

coding system; and computerized checklists guided the clinicians through some key safety practices like identifying the correct surgical site before the first incision. Moreover, about 50,000 patients had signed up to access a new electronic portal called MyChart, which allowed them to receive results of lab tests and x-rays, schedule appointments, refill their medications, and e-mail their physicians. Although there were grumbles here and there, the general feeling was that the electronic health record was making patient care safer and better.

Yet a series of dangers lurked beneath the placid surface. Installing a system like Epic is not like installing an operating system on your laptop, where you just “Accept the Terms,” reboot the machine and off you go. Instead, while the electronic health record provides the scaffolding, there are hundreds of decisions that each hospital needs to make, many of them related to electronic prescribing.

For example, should there be maximum dose limits set in the system, so that doses several times higher than the published maximum are grayed out? UCSF decided not to set such limits. The reasoning at the time was that, in a teaching hospital with lots of patients with rare diseases, many of them on research protocols, such “overdoses” would usually be okay. A system with hundreds of “hard stops” would lead to many angry phone calls from frustrated doctors to pharmacists, demanding that they override the block.



When it came to pediatric patients, a second set of decisions had to be made concerning weight-based dosing. Since pediatric patients can range from a preemie weighing a couple of pounds to a morbidly obese adolescent, many pediatric medications are dosed based on weight, usually expressed in milligrams (of a medication) per kilogram (of body weight) (mg/kg). The committee overseeing our Epic implementation decided to require weight-based dosing for all children under 40 kilograms (about 88 pounds).

Another choice involved the translation of weight-based doses into pills. What if the computer calculated that a dose should be 120 mg (based on the child's weight), but the only available pill was 100 mg? The decision: if the available medication was more than 5 percent off the calculated "correct" dose, then the pharmacist would contact the doctor to be sure she endorsed that conversion. After all, there might be cases in which a 10 or 20 percent disparity would be clinically meaningful and the doctor might rethink the order.

The weight-based dosing policy forced Lucca to order Pablo Garcia's medication in milligrams per kilogram, since the youngster weighed less than 40 kilograms (38.6 to be exact, or about 85 pounds). When she typed "Septra" into Epic's order entry module, she was prompted to select one of two dose choices, and she correctly chose the larger ("double-strength")

one, which contains 5 mg/kg of trimethoprim, one of the two active ingredients in Septra.

Because Pablo weighed 38.6 kilogram, the computer multiplied this weight by the 5 mg/kg and determined that the dose should be 193 mg.

sulfamethoxazole-trimethoprim (BACTRIM DS,SEPTRA DS) 800-160 mg tablet 160 mg of trimethoprim

Accept

Cancel

Link Order

Remove

5 mg/kg of trimethoprim × 38.6 kg = 160 mg of trimethoprim = 4.15 mg/kg of trimethoprim, Oral, Every 12 Hours Scheduled, First Dose Today at 2115

Reference Links:

1. Lexi-Comp

Dose:

5

mg/kg of trimethoprim

2.5 mg/kg of trimethoprim

5 mg/kg of trimethoprim

Weight Type:

Actual

Dosing

Order-Specific

Weight:

38.6 kg

Actual weight:

38.6 kg (recorded 11 hours ago)

⚠

160 mg of trimethoprim is the nearest dose that can be administered using available products (a decrease of 17% from the ordered dose of 193 mg of trimethoprim).

Administer Dose:

160 mg of trimethoprim

5 mg/kg of trimethoprim × 38.6 kg (Weight as of Tue Sep 10, 2013 0900)
= 193 mg of trimethoprim × 1 tablet/160 mg of trimethoprim
= 1 tablet × 160 mg of trimethoprim/tablet (rounded to the nearest 0.5 tablet from 1.2063 tablet)
= 160 mg of trimethoprim
= 4.15 mg/kg of trimethoprim

Administer Amount:

1 tablet

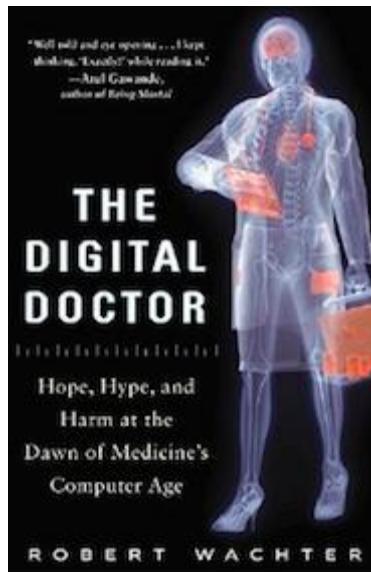
(rounded to the nearest 0.5 tablet from 1.2063 tablet)

Of course, there is no 193 mg Septra pill; the nearest tablet size is the 160 mg double-strength Septra pill. The computer recommended that the dose be rounded to a single tablet (a 17 percent decrease from 193, well above the 5 percent threshold), and asked Lucca if she accepted this recommendation.

She clicked “Yes.” In doing so, she believed she had ordered the one double-strength Septra tablet that Pablo had been taking at home, which was precisely what she had intended to do all along. All would have been well—had she been right. But she wasn’t.

Click here to read Part 2 of The Overdose





This is excerpted from The Digital Doctor: Hope, Hype, and Harm at the Dawn of Medicine's Computer Age, by Robert Wachter. McGraw-Hill, 2015. You can buy the book [here](#).

Part 2: Beware of the Robot

Pharmacist

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Part 3: Why

Clinicians Let Their Computers Make Mistakes

Part 4: Should

Hospitals Be More Like Airplanes?



Illustrated by *Lisk Feng*

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